

LUCIDMSK CME PROGRAM

MODULE 10 — POST-TEST

MSK Soft Tissue & Bone Tumors: A Systematic Imaging Approach

Soft Tissue Tumors · 10 Questions · 1.5 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq 70\%$ (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1

A 25-year-old with rapidly enlarging thigh mass. MRI shows intramuscular lesion with peripheral ossification and no enhancement. Most likely diagnosis:

- A Liposarcoma
- B Myositis ossificans (USP6 family)
- C Osteosarcoma
- D Lymphoma

2

BNCT diagnostic criteria (ALL required). Which combination is correct?

- A T1 high, T2 low, enhancement present, central vertebral
- B T1 low, T2 high, NO enhancement, central vertebral, $\pm$ mild sclerosis
- C Lytic lesion, T2 high, soft tissue mass, enhancement
- D T1 high, T2 low, no enhancement, eccentric cortex-based

3

Key MRI discriminating feature: lipoma vs. well-differentiated liposarcoma (WDLPS):

- A Size (>10 cm = sarcoma)
- B Deep vs. superficial location
- C Non-adipose (thick) internal septa or non-fatty nodular components
- D T1 signal of the fat component

4

Periarticular lobulated mass with low T1 AND low T2 signal in the knee. Most likely:

- A Ganglion cyst
- B Lipoma arborescens
- C PVNS / TSGCT
- D Baker cyst

5

Aggressive periosteal lesion in 15-year-old: Codman triangle + "sunburst" periosteal reaction + metaphyseal location. Most likely:

- A Ewing sarcoma
- B Conventional osteosarcoma
- C Osteoid osteoma
- D Giant cell tumor

6

Goutallier Grade 3 fatty infiltration in the gluteus medius. Most consistent with:

- A Normal aging
- B Greater trochanteric pain syndrome with chronic abductor tear
- C Acute gluteus medius strain
- D Piriformis syndrome

7

Typical BNCT vertebral lesion — correct management recommendation:

- A CT-guided biopsy immediately
- B PET-CT for staging
- C No biopsy — imaging surveillance appropriate; biopsy NOT recommended
- D Surgical resection

8

The "zonal phenomenon" of myositis ossificans on CT:

- A Dense central mineralization; lucent periphery
- B Peripheral mature mineralization; less organized soft center (outside-in pattern)
- C Uniform amorphous calcification throughout
- D Cortical disruption with central necrosis

9

Periosteal-based lesion arising from the posterior distal femur in a 35-year-old, WITHOUT intramedullary involvement. Most likely:

- | | |
|---|------------------------------------|
| A | Conventional osteosarcoma |
| B | Parosteal osteosarcoma (low-grade) |
| C | Periosteal chondrosarcoma |
| D | Giant cell tumor |

10

Most sensitive sequence for hemosiderin detection in PVNS/TSGCT:

- | | |
|---|---|
| A | T1 fat-sat |
| B | T2 fat-sat (STIR) |
| C | Gradient echo (GRE) — "blooming" artifact |
| D | Post-contrast T1 fat-sat |

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	B	B	C	C	B	B	C	B	B	C

Q1	A 25-year-old with rapidly enlarging thigh mass. MRI shows intramuscular lesion with peripheral ossification and no enhancement. Most likely diagnosis:	Answer: B
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A Liposarcoma

✓ **Myositis ossificans (USP6 family)**

C Osteosarcoma

D Lymphoma

EXPLANATION Myositis ossificans = USP6 family: young patient + intramuscular + peripheral-to-central ossification ("zonal phenomenon") + no enhancement of the ossified rim. Opposite pattern from osteosarcoma.

Q2	BNCT diagnostic criteria (ALL required). Which combination is correct?	Answer: B
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A T1 high, T2 low, enhancement present, central vertebral

✓ **T1 low, T2 high, NO enhancement, central vertebral, ± mild sclerosis**

C Lytic lesion, T2 high, soft tissue mass, enhancement

D T1 high, T2 low, no enhancement, eccentric cortex-based

EXPLANATION BNCT criteria: T1 hypointense, T2 hyperintense, NO enhancement, central vertebral, ± mild sclerosis, ± multifocal. Any frank lysis, extraosseous mass, or avid enhancement → raises chordoma. Biopsy NOT recommended for typical BNCT.

Q3	Key MRI discriminating feature: lipoma vs. well-differentiated liposarcoma (WDLPS):	Answer: C
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A Size (>10 cm = sarcoma)

B Deep vs. superficial location

✓ **Non-adipose (thick) internal septa or non-fatty nodular components**

D T1 signal of the fat component

EXPLANATION Key discriminator: non-fatty nodular components or thick (>2 mm) irregular septa = WDLPS until proven otherwise. Pure lipoma = thin (<2 mm) regular septa, homogeneous fat. Size and depth alone are insufficient.

Q4	Periarticular lobulated mass with low T1 AND low T2 signal in the knee. Most likely:	Answer: C
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A Ganglion cyst

B Lipoma arborescens

✓ **PVNS / TSGCT**

D Baker cyst

EXPLANATION PVNS / TSGCT = low T1 AND low T2 signal due to hemosiderin (paramagnetic). GRE sequences show blooming artifact. Ganglion = high T2. Lipoma = T1 bright. Baker cyst = T2 bright fluid.

Q5

Aggressive periosteal lesion in 15-year-old: Codman triangle + "sunburst" periosteal reaction + metaphyseal location. Most likely:

Answer: B

A Ewing sarcoma

✓ **Conventional osteosarcoma**

C Osteoid osteoma

D Giant cell tumor

EXPLANATION Osteosarcoma: Codman triangle + sunburst/spiculated periosteal reaction + metaphyseal location in adolescent. Ewing = diaphyseal + onion-skin periosteal reaction. Both may show Codman triangle.

Q6

Goutallier Grade 3 fatty infiltration in the gluteus medius. Most consistent with:

Answer: B

A Normal aging

✓ **Greater trochanteric pain syndrome with chronic abductor tear**

C Acute gluteus medius strain

D Piriformis syndrome

EXPLANATION Goutallier Grade 3 (fat = muscle) in the gluteus medius indicates chronic atrophy from long-standing abductor tendon tear — the "rotator cuff of the hip." Hallmark of chronic GTPS with established tendon failure.

Q7

Typical BNCT vertebral lesion — correct management recommendation:

Answer: C

A CT-guided biopsy immediately

B PET-CT for staging

✓ **No biopsy — imaging surveillance appropriate; biopsy NOT recommended**

D Surgical resection

EXPLANATION Typical BNCT does NOT require biopsy. Biopsy may miss the diagnosis or cause morbidity. Surveillance is recommended. Atypical features (lysis, soft tissue mass, enhancement) → atypical notochordal tumor → biopsy warranted.

Q8

The "zonal phenomenon" of myositis ossificans on CT:

Answer: B

A Dense central mineralization; lucent periphery

✓ **Peripheral mature mineralization; less organized soft center (outside-in pattern)**

C Uniform amorphous calcification throughout

D Cortical disruption with central necrosis

EXPLANATION

Myositis ossificans zonal phenomenon: mature peripheral ossification (organized cortical bone) with immature soft center — OUTSIDE-IN pattern. Osteosarcoma is the opposite (central matrix, disorganized periphery). Diagnostic on CT.

Q9

Periosteal-based lesion arising from the posterior distal femur in a 35-year-old, WITHOUT intramedullary involvement. Most likely:

Answer: B

A Conventional osteosarcoma

✓ **Parosteal osteosarcoma (low-grade)**

C Periosteal chondrosarcoma

D Giant cell tumor

EXPLANATION

Parosteal osteosarcoma: posterior distal femur (90%), periosteal-based (NOT intramedullary), low-grade, densely ossified, broad base on CT. Conventional osteosarcoma is intramedullary. GCT is epiphyseal/subarticular.

Q10

Most sensitive sequence for hemosiderin detection in PVNS/TSGCT:

Answer: C

A T1 fat-sat

B T2 fat-sat (STIR)

✓ **Gradient echo (GRE) — "blooming" artifact**

D Post-contrast T1 fat-sat

EXPLANATION

GRE sequences are most sensitive for hemosiderin — "blooming" artifact amplifies local field inhomogeneity. Hemosiderin is suppressed on spin-echo sequences. This is the hallmark of PVNS/TSGCT and hemophilic arthropathy.